

Week 1: EDA on MS MARCO

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1. Objective

Sized the dataset enough to make sampling decisions for later weeks. Specifically: how long are queries? How long are passages? What kinds of queries does MS MARCO have, and what kinds of answers do they get?

2. Dataset

- **Source:** HuggingFace `ms_marco` v2.1, **validation** split (101k queries) — chosen instead of `train` to keep the download light.
- **Sample:** first 5,000 rows of validation, deterministic.
- **Note:** this is the *QA* flavour of MS MARCO (queries + 10 candidate passages per query + a human-written answer + a `query_type` label). The Week 2 retrieval baseline uses the *Passage Ranking* corpus (8.8M passages, separate dataset).

3. Findings

3.1 Query length

Query length

Most queries are short (5–10 words). The long tail beyond ~15 words is small.

3.2 Passage length

Passage length (clipped at 200)

Passages are roughly 10× longer than queries (median ~50 words). Some passages are truncated to a single sentence — likely an artefact of the MS MARCO snippet extraction.

3.3 Query types

Query type distribution

DESCRIPTION and NUMERIC are the most common. LOCATION and PERSON are smaller buckets. The label is a noisy heuristic, not a clean taxonomy.

3.4 Answer types per query type

Answer types per query type

DESCRIPTION queries skew toward long answers; NUMERIC queries skew toward short / single-word. A meaningful fraction across all buckets is "no_answer" or empty, which we filter out before any supervision.

4. Implications for later weeks

- **Short queries** → **BM25 is reasonable as a first-stage retriever** (small vocab, lots of lexical signal). Confirmed by the Week 2 baseline.
- **Passage lengths < model context** (T5-small max input 512 tokens) → we can fit ~3 passages in the RAG generator's prompt without per-passage truncation in the common case.
- **no_answer and empty answers are common** → must be filtered for any fine-tuning supervision, otherwise the model learns to refuse.

5. Limitations

- 5k-row sample of the validation split. Exact percentiles will differ if you swap to train or to the full validation split.
- `query_type` is a coarse, noisy label. Not a basis for stratified evaluation on its own.
- HuggingFace `ms_marco v2.1` is the QA flavour; the figures here say little about the *retrieval* corpus, which is shaped differently.

6. Next

- Week 2: BM25 retrieval baseline on the official Passage Ranking corpus.
- Week 3: RAG generation conditioned on Week 2 retrievals.